

PROJECT WORKFLOW (END-TO-END)

1. USER ONBOARDING FLOW

Customer / Driver / Enterprise / Admin

1. User opens app/web portal
 2. Enter phone/email → OTP verification
 3. JWT authentication generated
 4. Profile setup (name, address, etc.)
 5. Role assigned (Customer / Driver / Enterprise / Admin)
 6. User lands on dashboard/home screen
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2. CUSTOMER BOOKING WORKFLOW

Booking Creation Flow

1. Customer selects:
 - Pickup location
 - Drop location
 2. System fetches distance via Google Maps API
 3. Customer selects:
 - Vehicle type
 - Goods type
 - Weight
 4. Backend calculates **dynamic price**
 5. Price estimate shown to customer
 6. Customer confirms booking
 7. Booking created in system (status: **pending**)
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3. DRIVER ALLOCATION WORKFLOW

1. Backend triggers **driver allocation algorithm**
 2. Nearby drivers fetched (based on GPS + vehicle type)
 3. Order pushed to drivers via:
 - Socket.IO
 - FCM notifications
 4. Driver receives request with timer
 5. Driver:
 - Accept → Assigned
 - Reject → Sent to next driver
 6. If accepted:
 - Booking status → **accepted**
 - Driver assigned to order
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4. LIVE TRACKING WORKFLOW

1. Driver app sends GPS every **3–5 seconds**
 2. Backend (Socket.IO):
 - Receives coordinates
 - Updates last location
 3. Broadcasts location to:
 - Customer app (real-time)
 4. Customer sees:
 - Moving driver on map
 - ETA updates
 - Status timeline
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5. ORDER EXECUTION WORKFLOW

Delivery Lifecycle

1. Driver navigates to pickup location
 2. Driver updates status:
 - Arrived
 - Picked Up
 3. Starts delivery:
 - Status → **In Transit**
 4. Reaches destination
 5. Delivery confirmation via:
 - OTP OR
 - Photo Proof (POD upload to S3)
 6. Status → **Delivered**
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6. PAYMENT WORKFLOW

Customer Payment Flow

1. Customer selects payment method
 2. Razorpay order created
 3. Customer completes payment
 4. Backend verifies:
 - Payment signature (webhook)
 5. Payment status updated:
 - unpaid → paid
 6. Invoice generated
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7. ENTERPRISE WORKFLOW (B2B)

Enterprise Booking Flow

1. Enterprise uploads:
 - CSV bulk orders OR manual entries
2. Backend validates data
3. Bulk bookings created
4. Orders follow same lifecycle as B2C

Billing Workflow

1. Monthly aggregation of orders
 2. Invoice generated (PDF)
 3. GST calculation applied
 4. Invoice sent via:
 - Email
 - Dashboard download
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8. ADMIN WORKFLOW

Operations Control

1. Admin monitors:
 - Active orders
 - Driver locations (live map)
 2. Can:
 - Assign/reassign drivers
 - Suspend drivers
 - Manage users
 3. Configure:
 - Pricing engine
 - Vehicle rates
 4. View analytics:
 - Revenue
 - Delivery success
 - Fleet utilization
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9. NOTIFICATION WORKFLOW

1. Trigger events:
 - Booking confirmed
 - Driver assigned
 - Out for delivery
 - Delivered
2. Backend sends:
 - Push notifications (FCM)
 - SMS (OTP)
 - Email (invoice)

10. SUPPORT & ISSUE WORKFLOW

1. User raises ticket from app
 2. Ticket stored in backend
 3. Admin responds via dashboard
 4. Notification sent to user
 5. Ticket marked resolved
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11. DATA FLOW (SYSTEM LEVEL)

End-to-End Data Movement

Client Apps (Flutter / Web)

REST API (HTTPS)

Node.js Backend (Services Layer)



- MongoDB (data storage)
- Redis (optional caching)
- AWS S3 (files)



External Services

- Google Maps (distance, tracking)
 - Razorpay (payments)
 - Firebase (notifications)
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12. COMPLETE ORDER LIFECYCLE (SUMMARY)

User Login



Create Booking



Price Estimate



Confirm Booking



Driver Assigned



Driver Accepts



Pickup → Transit → Delivery



Payment Completed



Invoice Generated



Order Closed
